

Greater Boston Area, MA

Rachel Hoffman-Bice

(850)418-3077

LinkedIn: <https://www.linkedin.com/in/rachelhoffmanbice/>

rhoffmanbice@proton.me

Website: <https://hobi362.github.io/>

Machine Design and Robotics Engineer seeking full time position focusing on the development, implementation and production-readiness of control systems, electro-mechanical interfaces and mechanical system design and development.

RELEVANT SKILLSET: Creo, Solidworks, MATLAB, Python, Git, Simulink, ROS, GD&T ASME Y14.5, IEC 60601, SAFe Agile, Jira, Linux OS, Raspberry Pi, OpenCV, Arduino, Injection Molding, Rapid Prototyping, Deterministic and Exact Constraint Design, DFM/DFA,

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Doctor of Philosophy in Mechanical Engineering, Minor in Leadership and Management, GPA: 4.9/5.0

June 2021

Dissertation: *Precision Assembly of Underconstrained Heavy Shafts Suspended By Multiple Cables From A Robotic Crane*

Master of Science in Mechanical Engineering, GPA: 4.9/5.0

June 2017

Thesis: *A Multi-Track Elevator System for E-Commerce Fulfillment Centers*

University of Florida, Gainesville, Florida, GPA: 3.97/4.0

May 2015

Bachelor of Science in Mechanical Engineering, Minor in Electrical Engineering, *summa cum laude, Outstanding Scholar Award*

PROFESSIONAL EXPERIENCE

Medtronic Surgical Robotics – Robotic Arm Hardware and Software Subject Matter Expert

February 2022 -Present

Senior Mechanical Design Engineer – New Product Development

May 2024 -Present

- Serve as Robotic Arm and Setup Arm (RSA) Hardware and Software Subject Matter Expert across the Surgical Robotics organization, leading design decisions, field failure investigations, testing and validation of both subsystems, with particular focus as hardware SME to multiple Hugo software release programs
- Technical lead for the next generation Control Arm (Insertion Axis (IA) and Instrument Drive Unit (IDU)) - own requirements definition, technical drawings per ASME Y14.5 and DFMEA-driven mechanical design, including tolerance stack analysis, FEA, loading conditions calculations, actuator performance, and validation against IEC 60601 and ANSI/AAMI HE75 requirements
- Coordinate Control Arm integration with software, electrical, operations, supply chain and human factors, and represent the Control Arm in project planning, with product development leadership and in the leading of technical design reviews
- On a prior next generation platform, held technical ownership of the IA primary actuator, IDU emergency release, custom IDU brake development and navigation hardware to IEC 60601 requirements, from concept to prototype delivery
- Mechanical Technical Lead of a Robotic Arm actuator field brake failure Corrective and Preventative Action (CAPA), setting the investigation strategy and authoring comprehensive engineering reports including failure cause mapping, tolerance stack analysis, component change impact assessments with vendor testing, and failure cause map analysis; closed with fix implemented in field

Senior Controls Software Engineer

2023-May 2024

Controls Software Engineer II

February 2022-January 2023

- Serve as Robotic Arm and Setup Arm (RSA) software technical expert and Product Owner for the Surgical Robotics organization, authored and maintained documentation for the design, testing and validation required for the RSA software release, including 37 software items, 23 requirements and 18 design documents released into Agile, and led the team through maintenance and execution of 77 verification test case
- Implemented safety workflows, joint validation functions and operating-mode updates (Manual Control, Teleoperation, Initialization) for a robotic arm with 6 actuated joints (one linear), 4 passive setup joints and a 4-DOF Instrument Drive Unit
- Authored software requirements and partnered with quality engineers to derive the DFMECA (per IEC 62304 and ISO 14971) and confirm software mitigations for identified risks were implemented, tested and captured in the design history file
- Integrated a Wristed Ligasure instrument plant model into a simulator to enable Model-in-the-Loop testing without hardware, and used system identification to develop a custom jaw controller

Medtronic GYN Minimally Invasive Therapies Group, Engineering Contingent

June 2021-February 2022

- Extensive usage of Solidworks to develop ASME Y14.5 compliant assembly drawings for export to suppliers for manufacturing
- Conducted verification testing on minimally invasive hysteroscopes to ensure resection performance met design specifications

MMID Full Service Design Team, Delft, The Netherlands, Functionality Design Intern

May 2017-Aug-2017

- Extensive concept development of a high-pressure pipe coupling, an apparatus for testing chemical dilution and a versatile electric outlet wall box, and a wall mounted broom clamp

MIT Lincoln Laboratory, Systems and Analysis Group, Mechanical Engineering Intern

May 2014-Aug-2014

- Evaluated and tested strategies to develop realistic optimal DTED terrain following aircraft trajectories with the goal of minimizing flight time and subject to velocity and acceleration constraints using the MATLAB optimization toolbox

SELECT PUBLICATIONS (Total: 4 Journal and 2 Conference publications)

- **R. Hoffman** and H. H. Asada, "Precision Assembly of Heavy Objects Suspended With Multiple Cables From a Crane," in IEEE Robotics and Automation Letters, vol. 5, no. 4, pp. 6876-6883, Oct. 2020, doi: 10.1109/LRA.2020.3013845
- **Hoffman, R.**, and Asada, H., "A Multi-Track Elevator System for E-Commerce Fulfillment Centers", 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Vancouver, 2017. **Best Application Paper Award**